

Human-AI Framework to Investigate New Promising Oncological Radiotherapy Techniques

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Abstract.

In recent years, Artificial Intelligence (AI) has shown promise for accelerating discoveries and improving scientific advancement in the field of research on effective new cancer treatments. However, there is a lack of a framework that systematically exploits the potential of AI in research programs to study novel radiation therapy approaches, which require a strong interaction with human expertise. We propose a framework integrating AI into multiple stages of a research pipeline devoted to the clinical translation of new FLASH and mini-beam radiotherapy techniques, but flexible and adaptable to study other emerging radiotherapy modalities. We detail a workflow scheme highlighting the potential of AI in each key step, including advanced image analysis, multi-parameter correlation analysis, and predictive modeling of treatment outcomes. We also stress the importance of a comprehensive data platform to organize experimental data to foster AI-based analysis, considering the possibility of integrating AI tools into the platform to streamline research. Although AI enhances data processing speed and precision, the researcher's knowledge of physics and biology remains crucial for contextualizing AI insights. We aim to stimulate the research community on how merging human expertise with AI capabilities could accelerate advanced radiotherapy techniques.

Keywords. Cancer Treatment, FLASH Radiotherapy, Minibeam Radiotherapy, Imaging Analysis, Artificial Intelligence, Machine Learning, Deep Learning

1. Introduction

Nowadays, the Artificial Intelligence (AI) revolution has already permeated the medical research domain, impacting various aspects of healthcare. AI intervenes by processing large volumes of data and uncovering valuable insights using sophisticated algorithms. Especially, AI has achieved remarkable advancements in medical imaging analysis, becoming a key player in the diagnosis and treatment of diseases [1]. In radiation therapy, AI has made substantial contributions, from optimizing radiation dose delivery to improving quality assurance and treatment precision [2]. Integrating human expertise with AI efficiency can also revolutionize the clinical translation of new radiation thera-

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pies. It can streamline experimental data analysis, automate labor-intensive tasks [3,4,5], and uncover patterns and correlations that are difficult to detect manually. However, to the best of our knowledge, no standardized framework that systematically exploits the potential of AI in all steps of preclinical investigations on novel radiation therapy approaches exists. This could be particularly relevant to integrating and analyzing diverse information from experimental data, simulations, radiobiological hypotheses, and other heterogeneous sources. Managing these complexities requires the involvement of a multidisciplinary team, and AI could complement human experts. These latter need to input information into the workflow that aligns with prior knowledge of the problem, while AI should go further and identify correlations in a data-driven manner. By ensuring that ‘clean’ and well-considered information is entered into the dataset by humans, the risk of confounding and bias effects that AI may be susceptible to is minimized.

This paper aims to present a possible human-AI collaboration framework designed to investigate new potential radiotherapy techniques, enhancing both the efficiency and speed of traditional preclinical studies, accelerating the discovery, and enhancing the accuracy and robustness of the research process. In the following, we will describe this potential workflow that harnesses the advantages of AI at multiple stages in this domain, specifically, in image analysis (including microscopy, biological sample imaging, and *in vivo* imaging) and interpretation of experimental data to understand the biological mechanisms underlying novel radiotherapy techniques.

2. Research Studies to Investigate New Radiation Therapies

Oncological diseases have a profound social and economic impact, posing a significant global challenge for both scientific research and technological innovation. Radiotherapy is a key treatment modality for cancer, utilized in more than a quarter of all cases [6]. However, novel approaches are necessary to advance beyond traditional methods, focusing on innovative radiation delivery techniques to enhance clinical outcomes [7]. This requires preliminary research based on experiments of different natures to evaluate the efficacy and safety of promising techniques before translating them into clinical practice.

2.1. FLASH and mini-beam Radiotherapy

The ‘FLASH effect’ is an observed radiobiological effect consisting in achieving similar tumor control as in conventional radiotherapy while significantly reducing damage to healthy tissues [8,9]. This can be obtained in the FLASH regime, i.e., by administering the therapeutic radiation dose in a single session over an extremely short time (< 200 ms) and at an ultra-high dose rate (≥ 40 Gy/sec). Although the effect has been demonstrated, the clinical translation of this new radiotherapy technique is currently prevented by two main obstacles: the radiobiological mechanisms responsible for the FLASH effect have not been fully understood yet, and new devices need to be developed for delivering radiation with such characteristics. For this reason, preclinical studies based on a multidisciplinary approach involving dosimetric characterization of the beam, *in vitro* and *in vivo* radiobiological experiments, and simulation and data modeling analyses are required to effectively translate this promising radiotherapy technique into clinics [10].

Minibeam radiotherapy is another under-study promising irradiation technique that involves spatial fractionation of the radiation dose distribution, which appears to pre-

serve healthy tissues while maintaining the same therapeutic effectiveness on the tumor [11,12]. It consists in delivering a non-uniform spatial dose using sub-millimetric parallel beams (0.5–1 mm) spaced in 1–3 mm. Also for this case, several challenges must be addressed before clinical implementation. These include technological hurdles related to beam delivery and the still-unknown dependencies of the minibeam effect on specific beam parameters. If successfully translated into clinical practice, these effects could revolutionize radiotherapy, bringing substantial clinical, economic, and social benefits. Therefore, a multidisciplinary approach based on beam delivery, dosimetry, radiobiology, and modeling experiments is crucial in view of the clinical translation of both techniques.

3. Human-AI Comprehensive Framework

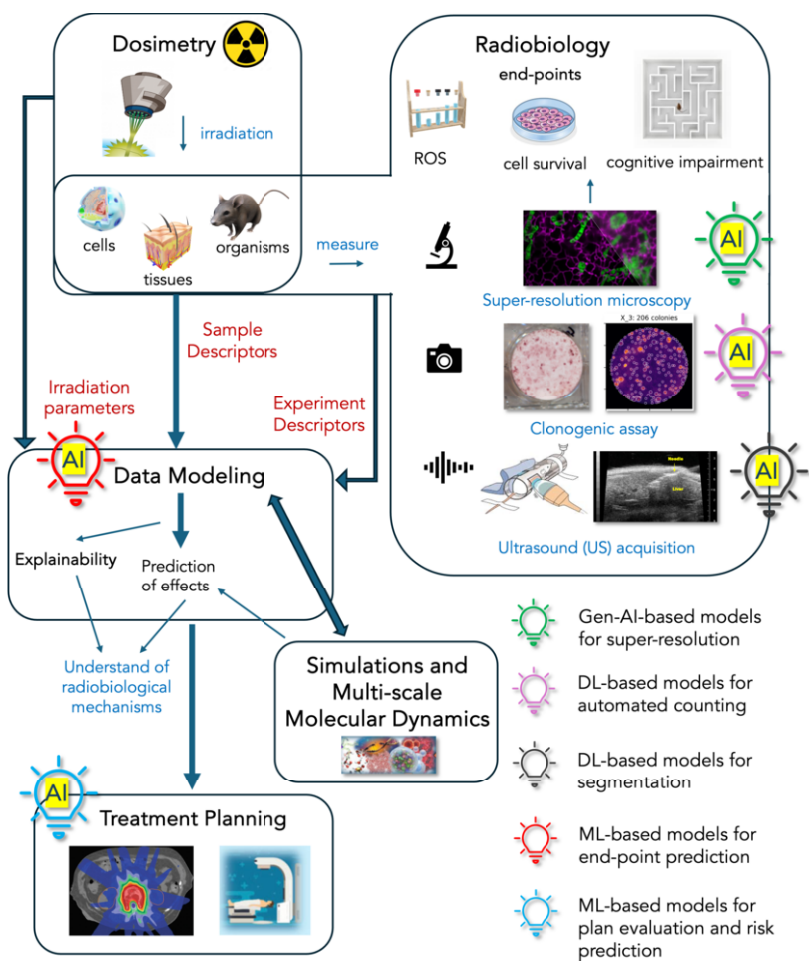


Figure 1. Scheme of the proposed human-AI preclinical study workflow to investigate new radiotherapy techniques, with reporting of where AI can intervene.

The combination of AI and human expertise is particularly promising in this research on new radiotherapy modalities, where the complexity of the intricate biological mechanisms at play requires both advanced computational tools for meaningful interpretation and human knowledge of physics and biology principles to ensure the reliability of the treatment [10]. It is essential to emphasize the need for a unified and standardized workflow in which the stages where AI can intervene are clearly defined, along with the specific points of interaction with human expertise.

To clarify the idea we want to present, we schematized the proposed workflow in Figure 1 and we explain in detail the individual AI-based steps and how they could be integrated within the overall research process in the following.

Actually, several parallel research programs are currently being developed worldwide to test different irradiation modalities (e.g. the use of different particles as protons or electrons to obtain the FLASH effect [13,14]; microbeam radiotherapy is also under investigation [15]; and the combined effect of FLASH and minibeam is being studied too [16]). Thus, the AI-based workflow that we propose could be re-used and re-adapted to all these similar investigations that require preclinical studies based on radiobiology experiments and data modeling. This framework, based on the collaboration between humans and AI, has the potential to revolutionize preclinical research in several ways, mainly by accelerating discoveries, increasing effectiveness, and reducing human error.

3.1. Deep Learning for Image Analysis

AI can intervene in different radiobiology experiments that are based on image acquisition and analysis to speed up the process. In Figure 1, we highlighted some main stages that can be expedited by Deep Learning (DL) tools. While AI can detect patterns and trends, human experts must act as the final validators and interpreters of results, using AI to guide and accelerate their exploration but not as autonomous decision-makers.

3.1.1. Generative AI for Super Resolution

Radiobiological experiments are primarily conducted by acquiring and analyzing microscopy images of the irradiated samples. When subcellular structures have to be observed, AI algorithms can be used to generate super-resolved microscopy images. Super-resolution microscopy (SRM) techniques enabling nanoscale observation exist. However, they suffer from long acquisition times, a significant limiting factor in radiobiology experiments. Thus, generative AI can help in generating super-resolution microscopy images from low-resolution ones, where the latter can be acquired with conventional fluorescence microscopy in a reasonable amount of time [17].

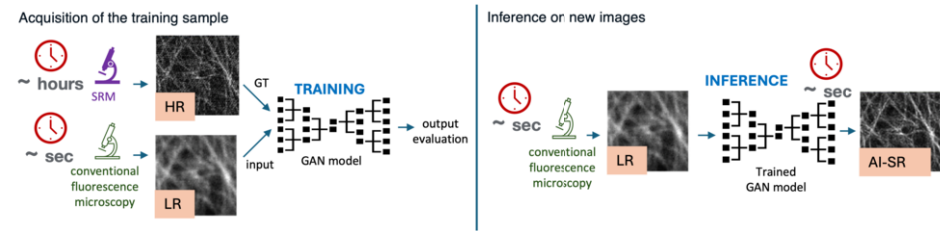


Figure 2. Generative AI-based workflow for generating super-resolution microscopy images. LR: Low Resolution, HR: High Resolution, AI-SR: Super Resolution image generated by AI.

In Figure 2, we schematize the designed workflow, starting from the acquisition of a reasonable amount of low-resolution and SRM high-resolution pair images to train the network, to the inference of the trained model able to generate the super-resolved image in the order of seconds. This could accelerate the investigation of numerous physiological and pathological processes at the subcellular level, including structural effects induced by radiation therapy. Nevertheless, the biophysicist researcher is responsible for interpreting the AI-enhanced visuals, using their expertise to contextualize observations. In particular, perceptual quality does not always correlate with commonly used AI metrics. Therefore, expert human validation is necessary to evaluate the quality of generated images.

3.1.2. Deep Learning for Colony and Cell Counting

Another task that can be automatized with a great saving of time and human effort is cell colony counting in images from clonogenic assay experiments. This experiment is an *in vitro* cell survival assessment based on the ability of a single cell to grow into a colony, where a colony consists of at least 50 cells. This is typically used to evaluate the effect of the irradiation on the sample. The quantification of the number of colonies is a step typically done by manually counting the colonies one by one, a really time-consuming and highly operator-dependent measurement.

This tedious and repetitive task could be accelerated by an AI algorithm properly trained on sample images to perform this counting automatically. The high variability of the sample images makes it difficult to apply a simple threshold-based method because colonies can appear differently depending on the observed cell line. Hence, a segmentation or clustering AI model trained on a large dataset could be of help in reducing analysis time and increasing accuracy [18]. In the workflow we propose, DL can streamline the process, but human oversight also remains crucial for this task. A pipeline with humans in the loop can be thought of to set the parameters, validate the outcomes, and correct any errors, ensuring the accuracy of the results.

This segmentation/counting task could also be translated to digital fluorescence microscopy images to identify and count the number of nuclei with a certain feature, such as with a specific grade of brightness or a particular shape. Currently, this task is performed mainly manually, but an AI model could accelerate the analysis.

3.1.3. Deep Learning for Tumor Segmentation in Ultrasound Images

In *in-vivo* experiments, an important measurement to evaluate the effect of specific irradiation on an animal model is the tumor volume quantification before and after irradiation. This requires an initial step of delineating the tumor volume on acquired ultrasound images of the animal, which is typically done manually and serves as the foundation for subsequent analysis [19]. According to the latest advancements, automated segmentation on medical images, and particularly ultrasound images, is widely explored with different models [20]. Therefore, this task could also be supported by AI, bringing a great advantage in the development of the experiment.

3.2. Machine Learning to Understand Complex Radiobiological Mechanisms

The study of the unknown radiobiological effects that drive these potential new radiotherapy techniques, such as FLASH or minibeam, is a multivariate problem. Multiple de-

pendent and independent variables affect the mechanisms, including both biological and physical variables that describe the irradiation (see Table 1) [21,22]. Therefore, a crucial and novel intervention of AI could be the development of clustering techniques or neural network models to identify patterns and reveal non-trivial associations and relationships among parameters that would not be revealed by traditional analysis.

Table 1. Some of the biological and physical parameters involved in preclinical experiments to investigate radiation therapy effects.

Bio in vitro	Cell line	Environment	Endpoint	Cell cycle	DNA content	...
Bio in vivo	Animal	Irradiated area	Body weight	Sex	Age	...
Radiation	Source	Particle	Charge	Energy	LET	...
Beam Structure	Dose total	Dose per pulse	Dose rate	# of pulses	Exposure time	...

A Machine Learning (ML) pipeline trained on *in vitro/in vivo* data can be used to predict a radiobiological endpoint (e.g., cell survival, mutation, etc.) [23]. The advantage of an AI-driven analysis is the possibility to apply explainability techniques to quantitatively understand the dependencies of the radiobiological effect on the beam and biological parameters and to study the impact of each parameter separately. The main challenge related to this approach could be defining a normalization and standardization strategy for the parameters because as clear from Table 1, these could be of different types (strings, categorical variables, continuous numerical variables in different ranges, etc.).

Different multiscale empirical models have been developed to date to describe the radiobiological response [24,25]. However, this approach, based on multiscale modeling of bio-processes, needs optimization of the parameters. This fitting task could be solved with an ML strategy by optimizing the model based on the experimental and simulation data. Thus, the speed and predictive capabilities of AI should be combined with the underpinnings of mechanistic biophysical models that simulate the biological processes to fast predict the effect on different tissues [26]. In this collaboration, the empirical knowledge earned from physics, chemistry, and biology observations remains fundamental, thus placing the role of the human researcher at the center of the workflow. Regarding this, we also suggest exploring Physics-Informed Neural Networks (PINN) and their extension to biological systems, which can incorporate physical and biological constraints into AI models, a novel application that could be investigated to further enhance the accuracy and interpretability of predictions in advanced radiotherapy research [27].

3.3. Machine Learning for Treatment Planning

ML can also be functional to the last step of a preclinical study, i.e., the biological optimization of treatment plans used in a final clinical context. A treatment plan is a patient-specific strategy designed to deliver radiation to a target while minimizing exposure to surrounding healthy tissues. It includes defining the radiation dose, the best parameters for beam delivery, and using imaging to ensure precise targeting.

The possibility of extracting mathematically defined features (radiomics or dosiomics features) from medical images, i.e. quantitative characteristics of the image that the human eye can miss, could be exploited to train an ML model on these features and predict the quality of a treatment plan or identify possible radiation-induced normal tissue toxicities [28,29]. Monitoring the late radiation toxicity can be discriminating in choosing one treatment plan over another.

4. The Crucial Role of a Data Platform

In the experiments described before, very heterogeneous data types are collected, from numerical parameters to tabular data, images, textual data, XML and raw data files. These data need to be interconnected to be properly analyzed. Studying the mechanisms underlying the biological effects of irradiation requires grasping complex relationships and associations among these data of different origins and typologies. Therefore, having a common and properly organized database to store, organize, share, and analyze these heterogeneous data is essential [30]. We suggest an IT platform that follows the schema in Figure 3, with sub-modules specific to the different experiments and properly configured to store the specific data, i.e., images, tabular data, end-point parameters, etc., related to the irradiation parameters used to irradiate the sample, and a main module in which the main parameters and data are shared. The final aim is to have a super-user that can access these data to perform correlation analysis through ML, simulations, and data modeling techniques to comprehend the radiobiological mechanism and then share the output to the final module devoted to treatment plan optimization.

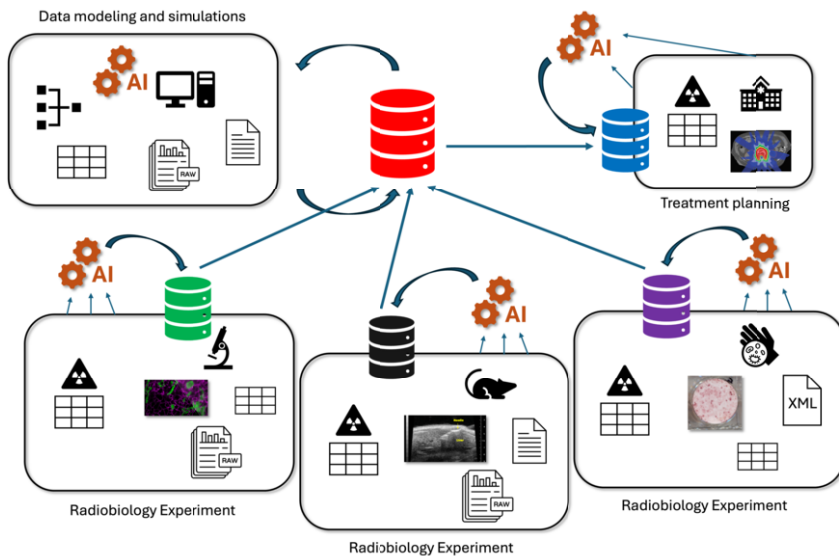


Figure 3. General schema of the experimental areas with the related data types to be stored and how they must be interconnected within the platform.

This kind of platform can be very useful to disentangle the different experiments that produce very heterogeneous data and facilitate the integration of AI-based tools in the described workflow. As schematized in Figure 3, the idea is to integrate the described AI tools within this platform so that they can directly elaborate the stored data and the output can be returned to the same database as an additional parameter. A scalable hardware architecture is essential for handling AI-driven image analysis efficiently. Processing data within the same server as the database limits scalability, making it necessary to interface with high-performance computing (HPC) resources. This is particularly crucial

for computationally intensive tasks, such as generative AI models for super-resolution microscopy, which may require at least an 8 GB VRAM GPU for inference only.

The human researchers must be involved in defining the relevant data to be included in the platform and how they are organized to facilitate the AI analysis. This centralized platform can also play a crucial role in improving data standardization and interoperability among various research institutions, undoubtedly accelerating clinical translation.

5. Conclusions

This paper aims to emphasize the need for a well-established framework for human-AI collaboration, where AI is seamlessly integrated into specific tasks to accelerate the clinical translation of new oncological radiation therapy techniques. Researchers from various multidisciplinary fields often express the need for tools that can speed up specific processes and enhance their knowledge. Although state-of-the-art AI methods already exist for many of these tasks, the lack of a structured workflow leaves researchers with limited guidance. Establishing a clear workflow would allow AI to act not just as a support tool but as an active ‘partner’ in this specific research field.

In the pilot study of the real-world implementation of the proposed framework, we outline a two-tiered evaluation approach. First, for the individual AI-driven components, expert-based validation will be conducted by comparing AI-generated outputs with conventional methods used by researchers. Second, the overall framework will be tested using existing datasets from completed preclinical studies to evaluate the ability of the pipeline to extract meaningful insights. However, a major open question remains: how to best validate ML models designed to enhance mechanistic understanding. Unlike predictive models, where performance can be assessed against known outcomes, AI-driven mechanistic insights require independent verification. Future work should explore robust methodologies for testing AI-generated hypotheses, ensuring their biological relevance.

Another challenge related to the effective implementation of the proposed framework is the availability of high-quality, large-scale datasets. As AI models require extensive and diverse training data to ensure robust performance, FLASH and mini-beam radiotherapy investigations are still in their early stages, and available data remains limited. For some of the AI-based tasks, such as image analysis, a possible solution is the use of preclinical data in other contexts for a first training of the models to then fine-tune them on data acquired in these new FLASH and Minibeam investigations. Additionally, variations in experimental conditions across different laboratories may introduce biases, complicating the generalizability of AI-driven insights. Standardization of protocols and data formats will be crucial to mitigate these discrepancies. The designed data platform could contribute to these much-needed standards for data types, parameters, and definitions.

Acknowledgements

The authors acknowledge the European Union - Next Generation EU, Mission 4 Component 2 Inv. 1.5, ECS00000017 ‘Ecosistema dell’innovazione’ Tuscany Health Ecosystem - THE project (Spoke 1 ‘Advanced Radiotherapies and Diagnostics in Oncology’), CUP I53C21000350006; INFN - CSN5 under the Minibeam Radiotherapy (MIRO) and Artificial Intelligence in Medicine: focus on Multi-Input Analysis (AIM_MIA) projects.

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